



ADDRESS

RMZ Millennia, Chennai, Tamil Nadu,
600096



CONTACT

+91-9043714778



EMAIL

contact@quntrolsphere.com

Custom FPGA Mezzanine Card (FMC) for ZCU102

Overview

The custom FPGA Mezzanine Card is a high-performance mezzanine module designed for the Xilinx UltraScale+ ZCU102 evaluation platform. The card enhances the ZCU102 with precision timing, high-speed signal interfacing, and flexible control I/O, thereby expanding the capabilities of the ZCU102 board to facilitate R&D experiments for Quantum Key Distribution (QKD), timing, and other security-focused applications.

Addressing Platform Gaps for Quantum Key Distribution (QKD)

Standard Xilinx UltraScale+ evaluation boards, including the ZCU102, are designed as general-purpose FPGA development platforms. While they provide high-performance processing and high-speed serial connectivity, they are not equipped with the full set of specialised I/O, timing, and control interfaces required for practical QKD system development and testing.

The Custom FMC Interface Card has been specifically developed to bridge this gap. By extending the ZCU102 with a reconfigurable, application-focused I/O and timing subsystem, the FMC card provides the missing hardware infrastructure needed to kickstart end-to-end QKD development, from early-stage experimentation to system-level testing and integration.

Key Features:

- HPC connector compliant with the ZCU102 interface
- Precision timing measurement with on-board TDC IC
- High-speed clock input for synchronising with an external device.
- High-speed clock output for the Single Photon Detector Gate signal.
- Electrical signal connectivity for controlling various MEMS-based electro-optical components, such as optical switches, variable optical attenuators etc.
- Photodiode monitoring circuitry for calibrating optical power.



QuntrolSphere

Pvt. Ltd.

CIN:U72100TN2025PTC181268



ADDRESS

RMZ Millennia, Chennai, Tamil Nadu,
600096



CONTACT

+91-9043714778



EMAIL

contact@quntrolsphere.com

Technical Description

FMC Standard	FMC High Pin Count (HPC)
Base Platform	Xilinx ZCU102
Clock I/O	SMA/SMB based
Control Signals	Trigger, gate, auxiliary I/O, etc

Designed for Reconfigurable QKD Development

The FMC card transforms the ZCU102 from a generic evaluation platform into a QKD-ready development system by offering:

- Dedicated interfaces for timing-critical and event-driven operation
- Flexible clock input/output and synchronisation capabilities
- FPGA-controlled trigger, gate, and auxiliary control signals
- Support for optical and electrical signal paths commonly used in QKD systems
- Firmware-defined behaviour that can evolve with protocol and system requirements

This reconfigurable approach ensures that the same hardware platform can support multiple QKD architectures, protocol variants, and experimental configurations without requiring board redesigns.

Support for SETS-Developed Key Distribution Engine (KDE)

The FMC has been architected for seamless integration with the Key Distillation Engine (KDE) development efforts at the Society for Electronic Transactions and Security (SETS), Chennai.

The FMC would allow for test stimulus generation for various interfaces necessary for the physical layer of QKD to enable KDE validation